Three-Tier Architecture Implementation in Microsoft Azure

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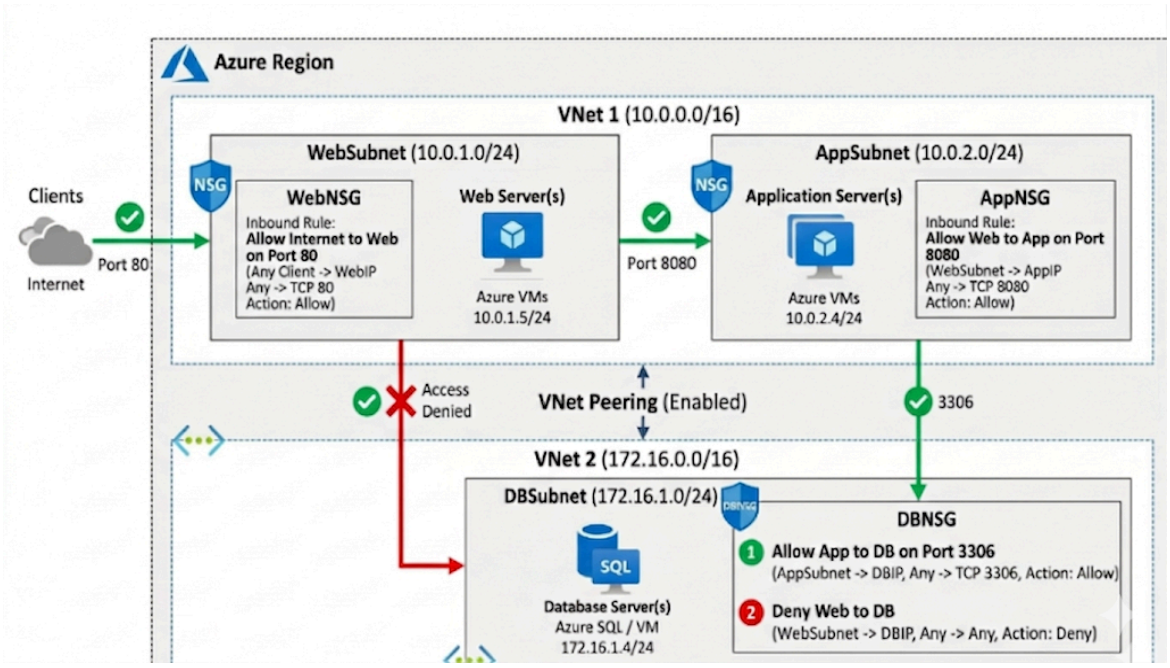
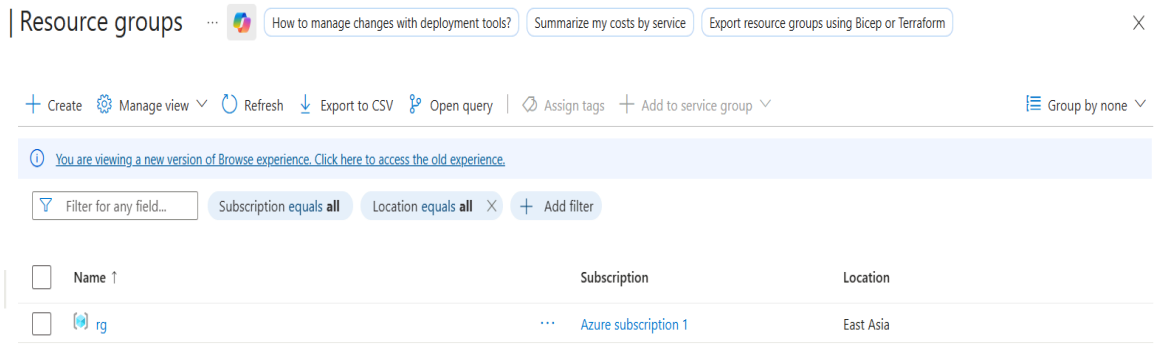
Overview:

This project implements a Three-Tier Architecture in Microsoft Azure to ensure secure, scalable, and modular application deployment. The architecture is divided into three logical layers: Web Tier, Application Tier, and Database Tier. Each tier is isolated within its own subnet and communicates through controlled network rules.

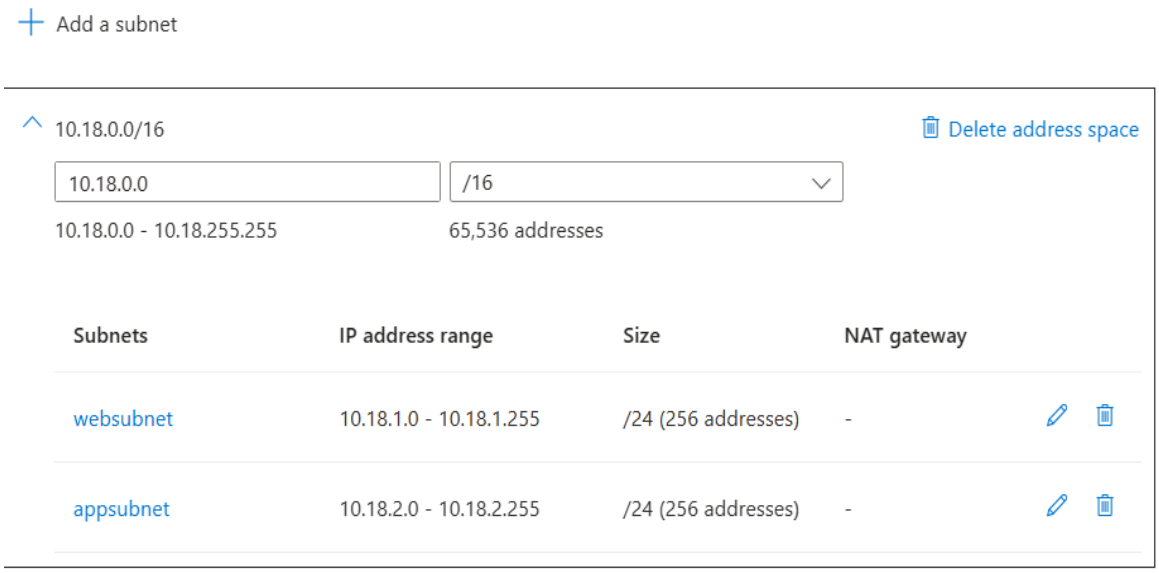
Tier	Component	Technology Used	Subnet	Access Type
Web Tier	Web Server	Nginx	Web Subnet	Public Access
Application Tier	Application Server	Apache Tomcat	App Subnet	Private Access
Database Tier	Database Server	MySQL	DB Subnet	Private Access



Step 1: Created a Resource Group (RG) in East asia region.
a) Selected Availability as No Infrastructure Redundancy and Security type as Standard.



Step 1: Created a VNet in the East-US region, and created two subnets within the same VNet for web and app.



Step 2: create web server virtual machine and select the websubnet,and allow the ssh port from any resource and destination as any and as for port 80- source is any and destination is our vm private ip in the nsg rules.

Webserver-vm

Virtual machines | Get started

Create | Reservations | Manage view | Refresh | Export to CSV | Open query | Assign tags | Start | Restart | Stop | ... | Group by none

You are viewing a new version of Browse experience. Click here to access the old experience.

Filter for any field... | Subscription equals all | Type equals all | Resource Group equals all | Location equals all | Add filter

Name ↑	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
webvm	Azure subscript...	rg	East US	Running	Linux	Standard_DC1s...	20.102.59.148	1

Webnsg rules:-

Network security group webvm-nsg (attached to networkInterface: webvm538)
Impacts 0 subnets, 1 network interfaces

Create port rule

Search rules | Source == all | Destination == all | Protocol == all | Action == all | Port == all

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (5)						
300	SSH	22	TCP	Any	Any	Allow
310	http	80	Any	Any	10.18.1.4	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (3)						

Step 4: now through ssh port we can connect to our VM and install our nginx web server.
From normal user - $ to admin user - #, command- sudo su
App update and apt install nginx -y
After checking our public IP to verify that our server is working.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Step 3:- create App server virtual machine and select the Appsubnet,and allow the ssh port (22) from source- webserver to destination- App server in the nsg rules, through the port (8080).

Virtual machines | Get started

Create | Reservations | Manage view | Refresh | Export to CSV | Open query | Assign tags | Start | Restart | Stop | ... | Group by none

You are viewing a new version of Browse experience. Click here to access the old experience.

Filter for any field... | Subscription equals all | Type equals all | Resource Group equals all | Location equals all | Add filter

Name ↑	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
appvm	Azure subscript...	rg	East US	Running	Linux	Standard_DC1s...	138.91.115.38	1
webvm	Azure subscript...	rg	East US	Running	Linux	Standard_DC1s...	20.102.59.148	1

Appserver nsg rules:

Network security group appvm-nsg (attached to networkInterface: appvm589)
Impacts 0 subnets, 1 network interfaces

Create port rule

Search rules | Source == all | Destination == all | Protocol == all | Action == all | Port == all

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (4)						
310	allow-tomcat	8080	Any	10.18.1.4	10.18.2.4	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (3)						

Install tomcat version to check app server works:

1) To login from web to app, follow the command- ssh username@privateip app vm and enter the password
2) Install java version if the application works on it - apt install default-jdk - y
3) Install tomcat10 - y.

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

admin18@appvm:~$
```

```
Get:56 http://azure.archive.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 40.8 MB in 5s (8514 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
28 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@appvm:/home/admin18# apt install default-jdk -y
```

webvm | 0% | 0.57 GB / 7.76 GB | 0.10 Mb/s | 0.06 Mb/s | 33 min | admin18 (x2) | /: 7% | /boot: 8% | /boot/efi: 6%

rt MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@appvm:/home/admin18# apt install tomcat10 -y
```

webvm | 0% | 0.57 GB / 7.76 GB | 0.02 Mb/s | 0.00 Mb/s | 35 min | admin18 (x2) | /: 7% | /boot: 8% | /boot/efi: 6%

DB-server:

Step 4: Create a VNet in another region for db server, within create a vm and nsg rules for it, and VNet-peering for both regions to connect.

Create | Manage view | Refresh | Export to CSV | Open query | Assign tags

Filter for any field... | Subscription equals all | Resource group equals all | Location equals all | Add filter

Showing 1 to 2 of 2 records. | No grouping | List view

Name	Resource group	Location	Subscription
vnet-1	rg	East US	Azure subscription 1
vnet-2	rg	West US	Azure subscription 1

Virtual machines | Get started

Create | Reservations | Manage view | Refresh | Export to CSV | Open query | Assign tags | Start | Restart | Stop | Group by none

You are viewing a new version of Browse experience. Click here to access the old experience.

Filter for any field... | Subscription equals all | Type equals all | Resource Group equals all | Location equals all | Add filter

Name	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
appvm	Azure subscript...	rg	East US	Running	Linux	Standard_DC1s...	138.91.115.38	1
dbvm	Azure subscript...	rg	West US	Running	Linux	Standard_DC1s...	20.237.223.245	1
webvm	Azure subscript...	rg	East US	Running	Linux	Standard_DC1s...	20.102.59.148	1

Virtual network peering enables you to seamlessly connect two or more virtual networks in Azure. The virtual networks appear as one for connectivity purposes. Learn more

Filter by name...

Showing all 1 items

Name	Peering sync status	Peering state	Remo...	Virtu...	Cross-tenant
vnet-2-1-peering	Fully Synchronized	Connected	vnet-1	Disabled	No

Network security group dbvm-nsg (attached to networkInterface: dbvm478)
Impacts 0 subnets, 1 network interfaces

Create port rule

Search rules | Source == all | Destination == all | Protocol == all | Action == all | Port == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (5)						
310	allow-app-db	3306	Any	10.18.2.4	100.0.1.4	Allow
320	deny	Any	Any	10.18.1.4	100.0.1.4	Deny
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (3)						

Step 5: From app to dbserver to login,

1) Type - ssh username@privateip and enter password- same method for any vm to login from their to another vm.

```
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '100.0.1.4' (ED25519) to the list of known hosts.
admin18@100.0.1.4's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1008-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Mar 22 04:43:52 UTC 2026

  System load:  0.0               Processes:             125
  Usage of /:   5.7% of 28.02GB   Users logged in:       0
  Memory usage: 4%                IPv4 address for eth0: 100.0.1.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

admin18@dbvm:~$
```

STEP 6 : To install mysql db, we have to follow the commands and the rules to edit

sudo apt update
sudo apt install mysql-server -y
sudo systemctl status mysql sudo mysql_secure_installation
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf

Find : bind-address = 127.0.0.1 Change To : bind-address = 0.0.0.0 sudo systemctl restart mysql sudo mysql -u root -p

```
Remove test database and access to it? (Press y|Y for Yes, any other key for No) : yes
 - Dropping test database...
Success.

 - Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : yes
Success.

All done!
root@dbvm:/home/admin18# sudo nano /etc/mysql/mysql.conf.d/mysql.cnf
root@dbvm:/home/admin18# nano /etc/mysql/mysql.conf.d/mysql.cnf
root@dbvm:/home/admin18# exit
exit
admin18@dbvm:~$ sudo nano/etc/mysql/mysql.conf.d/mysql.cnf
sudo: nano/etc/mysql/mysql.conf.d/mysql.cnf: command not found
admin18@dbvm:~$ sudo su
root@dbvm:/home/admin18# nano /etc/mysql/mysql.conf.d/mysqld.cnf
root@dbvm:/home/admin18# systemctl restart mysql
root@dbvm:/home/admin18# systemctl status mysql
● mysql.service - MySQL Community Server
     Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-03-22 05:25:49 UTC; 23s ago
    Process: 3033 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
   Main PID: 3043 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 9504)
     Memory: 362.7M (peak: 380.5M)
        CPU: 586ms
     CGroup: /system.slice/mysql.service
             └─3043 /usr/sbin/mysqld

Mar 22 05:25:48 dbvm systemd[1]: Starting mysql.service - MySQL Community Server...
Mar 22 05:25:49 dbvm systemd[1]: Started mysql.service - MySQL Community Server.
root@dbvm:/home/admin18#
```

Enter MySQL and follow the command below:

```
ot'@'localhost' IDENTIFIED WITH mysql_native_password BY '[redacted]' at line 1
mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY '[redacted]';
Query OK, 0 rows affected (0.03 sec)

mysql>
```

The above password works only inside db server where we are accessing the MySQL database.

So we are creating a new password that can connect appserver to mysqldb:

```
ysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY 'Hydnvs@8';
uery OK, 0 rows affected (0.03 sec)

ysql> CREATE USER 'appuser'@'10.18.2.%'IDENTIFIED BY 'App88@88';
uery OK, 0 rows affected (0.04 sec)

ysql> GRANT ALL PRIVILEGES ON *.* TO 'appuser'@'10.18.2.%';
uery OK, 0 rows affected (0.01 sec)

ysql> FLUSH PRIVILEGES;
uery OK, 0 rows affected (0.01 sec)

ysql>
```

Exit

Using the ssh username@privateip of the appserver, and enter, apt update and install mysql client:

```
28 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@appvm:/home/admin18# apt install mysql-client -y
Reading package lists... Done
```

TEST FROM APP SERVER: mysql -h db_private_ip -u appuser -p and enter the password that we gave for appuser, and after installation exit, and tested database connectivity using: telnet 3306

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@appvm:/home/admin18# mysql -h <100.0.1.4> -u appuser -p
bash: 100.0.1.4: No such file or directory
root@appvm:/home/admin18# mysql -h <DB_100.0.1.4> -u appuser -p
bash: DB_100.0.1.4: No such file or directory
root@appvm:/home/admin18# mysql --version
mysql  Ver 8.0.45-0ubuntu0.24.04.1 for Linux on x86_64 ((Ubuntu))
root@appvm:/home/admin18# mysql -h 100.0.1.4 -u appuser -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
webvm | 0% | 0.59 GB / 7.76 GB | 0.02 Mb/s | 0.01 Mb/s | 177 min | admin18 (x2) | /: 7% | /boot: 8% | /boot/efi: 6%
```

```
caching_sha2_password2#08S01Got timeout reading communication packetsConnection closed by foreign host.
root@appvm:/home/admin18# Hyderabad1234
Hyderabad1234: command not found
root@appvm:/home/admin18# telnet 100.0.1.4 3306
Trying 100.0.1.4...
Connected to 100.0.1.4.
Escape character is '^]'.
[
8.0.45-0ubuntu0.24.04.1
++q@l▒X~z)caching_sha2_password2#08S01Got timeout reading communication packetsConnection closed by foreign host.
root@appvm:/home/admin18#
```

END: In this project, a 3-tier architecture was successfully deployed in Azure with proper network segmentation and security. Web, Application, and Database tiers were isolated using subnets and secured using NSGs. MySQL database connectivity was established securely using a dedicated user instead of root access. This project demonstrates best practices in designing scalable and secure cloud-based applications.